Data request: the fall-degree paper (treatise Ch. 6 core)

2026-10-29

Purpose

This is a list of the data the paper still needs, one item per gap. Each item names:

- the claim it supports;
- what is on record now;
- what is missing;
- how to get it: a new run under a protocol record in `protocols/`, collecting output from a study already running, or work that needs a person (no compute).

Status (2026-10-30). The owner approved building the prerequisites and launching the runs.

- **The two fall-degree protocols** now use one native instrument, `examples/fd_m2_degrees.rs`.
  - It builds the $m = 2$ descent in crypto's Boolean-ring library. It runs the full MutantXL closure ($= W_D$) and Boolean F4.
  - It replaces the draft's lfdclose2 and a new `.sys` driver. crypto forbids Python research tooling, and the 120 GiB runner serves only this repository.
  - Pilot: it reproduced lfdclose2's recorded d_F on 9 recorded instances. These are R11a, R13a, R15a, R19a, P17a, P21a, P21c and U33a (exact), and T33a (≥ 5, with a W_4 rank of 48,068 equal to the record).
  - Planted solvable controls never refuted up to $D = 5$.
- **The runs** go through `.github/workflows/fd-paper-degrees.yml` on the `crypto-eks-highmem` runner (Karpenter Spot on EKS, 16 CPU, 120 GiB). This is the route `dreg-a3-eks-highmem.yml` uses for closures.
  - EKS `isolab_k8s_run` takes argv jobs only from the harness host, which this session cannot reach (`research/prime_ic_program_20261029/fall_degree_relaunch/README.md` §1).
  - The calibrate stage (claim C0) must pass before any new row is read.
- **Certificates** (checked by `cert_check.py`) for the new $d_F \geq 5$ rows are a separate crypto-autoresearcher run. Until then those rows are reported as `closure_only`.

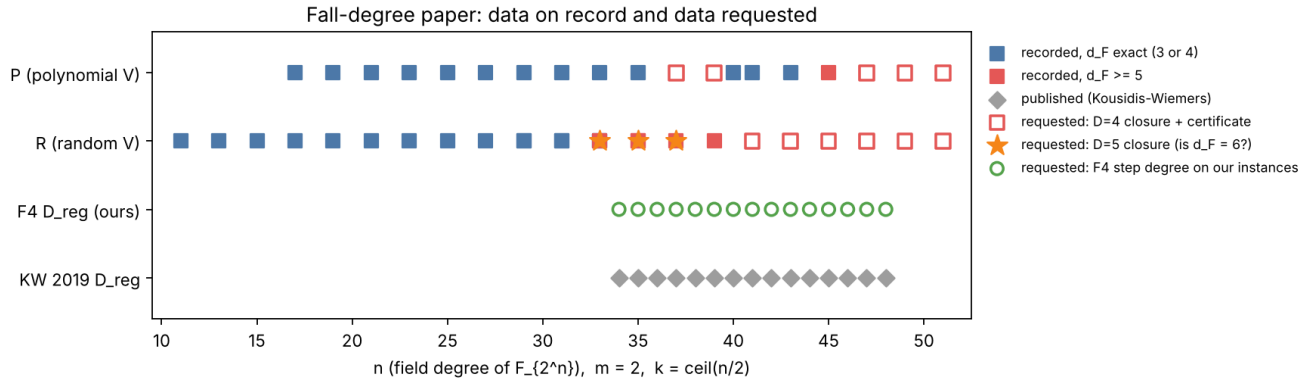


Figure 1: Data on record (filled) and requested (open, stars, circles) for the m = 2 binary descent

The paper's hypotheses and the data each needs

The paper proves:

- the first fall degree is a constant of the presentation: 2 for binary S_3 descent, 4 for PKM towers;
- the last fall degree is not bounded by it.

The hypothesis it cannot yet state is that **the last fall degree grows**. The table lists what would move that question, and what the paper's existing tables need to be complete and reproducible.

#	Claim in the paper	On record	Gap	How	Priority
R1	Thm 6.11: $d_F \geq 5$ certified (n = 33–45); Table 6.1	13 checked certificates; R39a/b and C21a/b closure A only	Certificates for R39a/b and C21a/b; random V at n = 41–51; polynomial V at n = 37, 39, 47–51	**FD-BINARY-CLOSURE-EXT**, claims C1, C3	1
R2	Open problem: does d_F reach 6?	R35a is exactly 5; no W_5 has ever been computed	Degree-5 closure on the 8 certified instances at n = 33–37	**FD-BINARY-CLOSURE-EXT**, claim C2	1
R3	Table 6.3: KW's D_{reg} against our d_F	KW published values only; no F4 step degree on our instances	D_{reg} on the same systems; reproduce KW at n = 34–48	**FD-F4-STEP-DEGREE**	1
R4	Table 6.4 and the C-step prediction $d_F = 4 + \lfloor (t-2)/4 \rfloor$	$t \leq 10$ medians session-reported, raw data lost	t = 1–14 re-run with committed rows; m = 4 rise/flat/fall reading	**Collect output** from the running relaunch: `research/prime_ic_program_20261029/fall_d` (Amendments 7–8). No new protocol.	1
R5	Every certificate "any reader can check"	sha256 values in `certificates.txt`; certificate files not public	Publish certificate files and `cert_check.py` with hashes	Aggregation: artifact store plus a public bundle	1
R6	Table 6.3 transcription; which subspace KW used	Read from a photographed page	Check against the PDF; note KW's subspace and z	Literature (person); read HKY 2015 and KW 2019 in full	1
R7	Closure times in Table 6.1	Host not recorded	Host recorded for new rows	Inside FD-BINARY-CLOSURE-EXT (instance type, CPU, peak RSS per row)	2
R8	Conclusion item 3: a measured crossover at k = 4	m = 15 only, n = 2 per arm	8 fresh targets per arm at m = 14–17, priced across the band	**K4-CHAR2-REPLICATION**	2
R9	PKM raw records from the first run	Lost on the owner's Mac (`RESTART.md`)	Upload the Mac worktree	Owner action	3

Out of scope for this paper: the frontier chapter's 9 unvalidated leaderboard rows. They are tracked by the validation gate, not here.

Protocols

protocol	question	arms and controls	size	route	prerequisites before freeze
`fd-binary-closure-ext`	Is $d_{\mathrm{alt}} \geq 5$ at n = 41–51? Does d_F reach 6 at n = 33–37?	native closure against lfdclose2's record (calibration C0); A/A on thread count; positive: recorded refuting instances; negative: planted solvable instances	93 recorded + 28 drawn + 10 planted + 8 D = 5; ~70 h	ARC `crypto-eks-highmem` 16 CPU, 120 GiB	done: driver, `run.sh`, workflow, pilot
`fd-f4-step-degree`	What is F4's D_{reg} on the same systems? Does it reproduce KW?	Boolean F4 (`pq_f4_f2`) against KW's published values; positive: d_F = 3 instances	93 recorded + 30 drawn; steps over 1 GiB are censored	ARC `crypto-eks-highmem`	done: driver (`run-f4`); still open: KW transcription check against the PDF
`k4-char2-replication`	At k = 4, is IC below the strongest rho at m = 16 and 17, across w = 0, 0.1, 1?	`ic-wied` against `rho-strong`; `rho-neg` regression; A/A	4 arms × 4 m × 8 targets; ~30 h	isolab L2 measurement pool	build `rho.negation_strong` (ecbench-extend); `ecbench verify --replay` preflight

Protocol details. Each record states:

- its claims, each with a class;
- its hypotheses, with dated predictions marked out-of-sample or calibration;
- a variable ledger in which every comparison is one-factor, absolute or A/A;
- the seed rule;
- the metrics, with exact or counted quantities as primary;
- stopping and invalidation rules;
- the verdict vocabulary;
- what is inadmissible.

Seeds. The k = 4 seeds are listed explicitly. The red team's plan wrote `sha256("gaudry-k4-redteam-v2" || m || i)` without a byte encoding. This record fixes it as UTF-8 `gaudry-k4-redteam-v2|<m>|<i>` before any data. The other two protocols derive their seeds from the protocol hash after freeze.

What comes back, and where it goes

Every run writes `experiment.export/v1` beside its records. Each result has a fixed destination:

- `check-export --verify-files` and the `flow` count go in the run report.
- New certified rows are appended to `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/results/`. `papers/ecdlp-index-calculus-treatise/figures/make_tab_fall_degrees.py` then regenerates Tables 6.1–6.3, with a measured D_{reg} column in 6.3.
- R4 output updates `figures/data-new.json` and Table 6.4.
- R8 reaches the paper only through the validation gate.

No outcome promotes a hypothesis by itself: review-evidence and the PR decide.